$\mathbb{R}^{2k}$ is Theoretically Large Enough for Embedding-based Top-k Retrieval

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1. DEFINITIONS

Setting: embedding-based retrieval

- **Universe:** $\mathcal{U} = [m] = \{1, \ldots, m\}$.
- **Object embeddings:** one shared configuration $v_1, \ldots, v_m \in \mathbb{R}^d$.
- **Score rule:** $s(v_i, u_S)$ ranks object i for query embedding u_S.
- **Embeddable (top-k retrieval view):**

$\boxed{\forall S \subseteq \mathcal{U},\ 1 \le |S| \le k}$

$\boxed{\exists \text{ one query embedding } u_S}$

$\boxed{\text{such that } i \mapsto s(v_i, u_S) \text{ distinguishes } S \text{ from } \mathcal{U} \setminus S}$

- **Minimal Embeddable Dimension (MED):** the smallest d for which one shared configuration in $\mathbb{R}^d$ distinguishes every target set $|S| \le k$.

Exact MED

$$\min_{i \in S} s(v_i, u_S) > \max_{j \notin S} s(v_j, u_S)$$

Strict ordering is enough. No prescribed margin or normalization is required.

ϵ-Robust MED

$$\min_{i \in S} s(v_i, u_S) \ge \max_{j \notin S} s(v_j, u_S) + \epsilon$$

Objects and queries are normalized, e.g. in unit ball.
A uniform positive score gap must work for every target set.

Fixed object embeddings; the witness query u_S changes with S.

VC lower bound

If all subsets of size at most k can be realized, then the thresholded score class shatters any chosen k objects. Thus

$$k \le \mathrm{VCD}(\mathcal{C}_{s,d}).$$

For inner-product and Euclidean threshold families, $\mathrm{VCD} = d + 1$, so exact MED and every robust MED satisfy $d \ge k - 1$.

Lower bounds are $\Omega(k)$; the question is whether m must enter.

2. THEORY

Exact capacity is m-independent; robust capacity pays for margin.

Cyclic-polytopal witness for Exact MED upper bound

Put all m objects on the moment curve in $\mathbb{R}^{2k}$:

$$v_i = (t_i, t_i^2, \ldots, t_i^{2k}), \qquad 0 < t_1 < t_2 < \cdots < t_m < 1.$$

For target S, consider the square polynomial $-P_S^2(t)$, where

$$P_S(t) = \prod_{i \in S} (t - t_i).$$

Note that P_S^2 has degree at most $2k$. Select u_S from the nonconstant coefficients of $-P_S^2(t)$, giving a constant shift c_S:

$$s(v_i, u_S) = \langle v_i, u_S \rangle = c_S - P_S^2(t_i) \begin{cases} = c_S, & i \in S, \\ < c_S, & i \notin S. \end{cases}$$

Exact MED theorem

Score	Lower	Upper
Inner product	$k-1$	$2k$
Euclidean	$k-1$	$2k$
Cosine	$k-1$	$2k+1$

Main message: $\mathrm{MED}(m, k) = \Theta(k)$, independent of m up to constants.

Feasible robust margin

$$\epsilon_\star(m, k) = \frac{m}{\sqrt{k(m-1)(m-k)}} \sim \frac{1}{\sqrt{k}}.$$

If $\epsilon > \epsilon_\star(m, k)$, robust retrieval is impossible in any finite dimension. The ceiling is tight in $\mathbb{R}^{m-1}$ by a regular simplex.

Gaussian centroid witness for robust MED upper bound

Sample random unit object vectors and query by normalized selected centroid:

$$u_S = \frac{\sum_{i \in S} v_i}{\left\| \sum_{i \in S} v_i \right\|_2}.$$

Selected objects get a self-correlation term; outsiders contribute coherence noise. With positive probability,

$$d = Ck^2 \log m$$

robustly retrieves every at-most-k subset with margin $\Omega(1/\sqrt{k})$.

RMED dimension ladder

At $\epsilon_k = c/\sqrt{k}$, packing and Gaussian witnesses give

$$\max\left\{ k - 1, \frac{\log \binom{m}{k}}{\log(1 + 2/\epsilon_k)} \right\} \le \mathrm{RMED}(m, k, \epsilon_k)$$
$$\le \mathrm{RMED\text{-}C}(m, k, \epsilon_k) \le Ck^2 \log m.$$

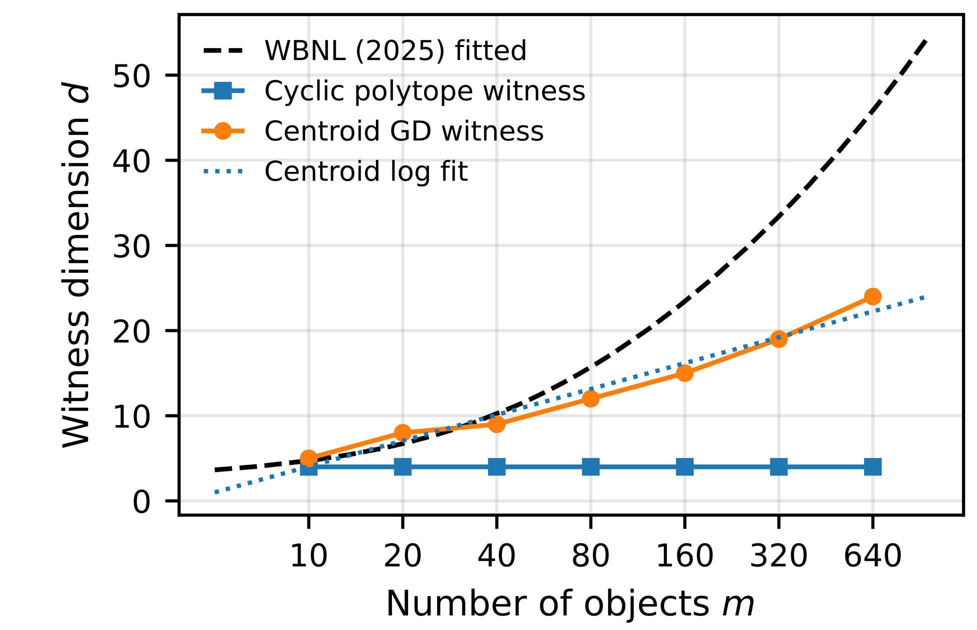
Exact MED supplies $k-1$; packing supplies the m-dependent term.

$2k$ is enough for exact MED; $O(k^2 \log m)$ is enough for robust MED.

3. EMPIRICAL

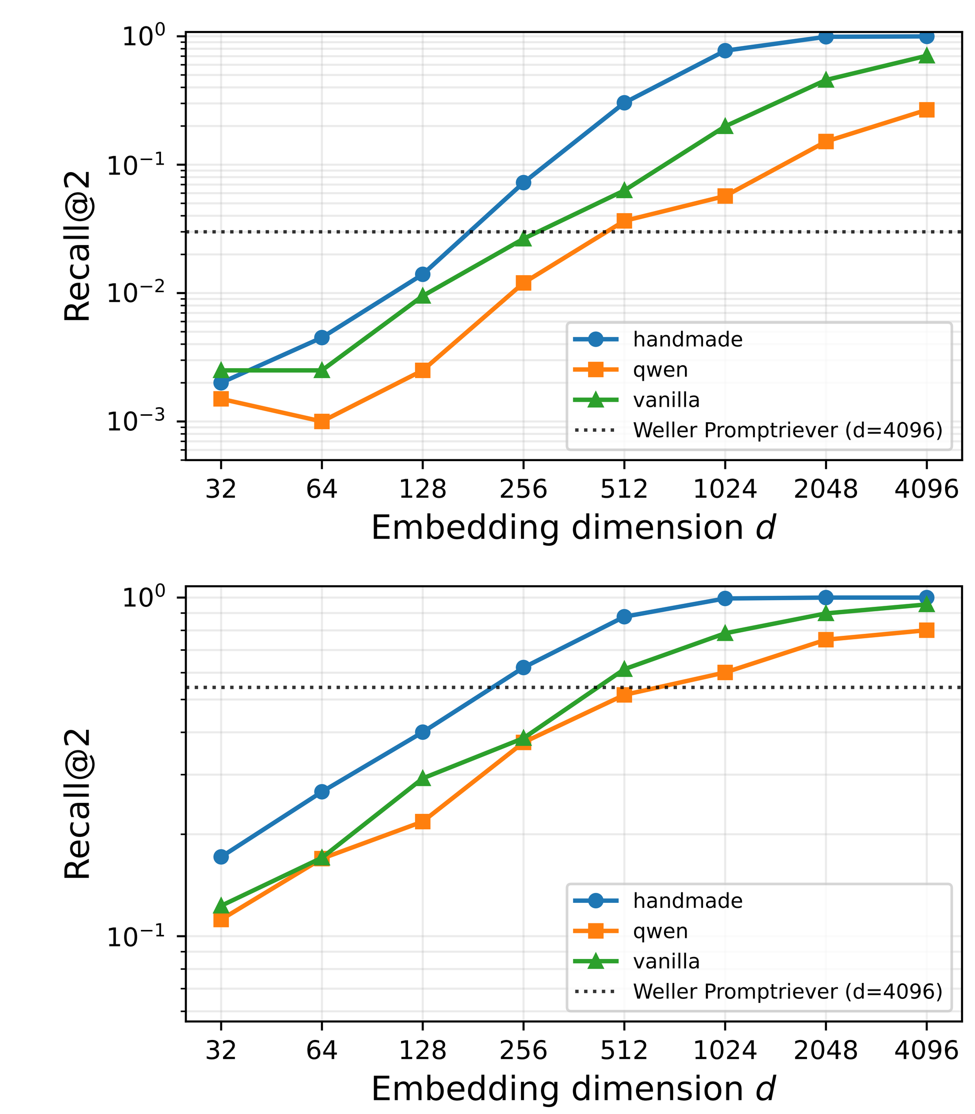
Current "hard" datasets are easy to crack by explicit low-dimensional witnesses.

Synthetic top-2: explicit $d = 4$



At $m = 640$, the cyclic construction checks $205120/205120$ singleton/pair queries.

LIMIT: random single-vector controls pass



Dataset	baseline first	vanilla d	vanilla R@2 @4096
LIMIT	0.030	512	0.7060
LIMIT-small	0.543	512	0.9545

Vanilla token sums use no labels and no handcrafted phrases, yet exceed the strongest comparable single-vector Promptriever line. Handmade reaches R@2 $0.9980/1.0000$ at $d = 4096$.

Takeaway:
Failures of embedding models DO NOT stress geometric capacity.